

AADK 3 – TASK 1 - Text Composable

Understanding & Exploration

Student Name: Rajat Prakash Dhal

Course/Unit: Android Development with Kotlin – Session3

USN: 1hk22cs115

Email: 1hk22cs115@hkbk.edu.in

1 Definition: What is a @Composable function?

A **@Composable function** in Jetpack Compose is a special function used to define UI elements in Android. It describes how a part of the screen should look and behave. Instead of using XML layouts, composable functions let us build UI directly in Kotlin code in a declarative way.

2 Three Key Text Composables

1. Text

- **What it does:**
Displays text on the screen.
- **Example use-case:**
Showing a heading like “Welcome” or displaying a paragraph in a blog app.

2. Column

- **What it does:**
Arranges UI elements vertically (one below another).
- **Example use-case:**
Stacking a title, subtitle, and paragraph from top to bottom.

3. Row

- **What it does:**
Arranges UI elements horizontally (side by side).
- **Example use-case:**
Placing two text buttons like “Login” and “Register” next to each other.

◆ How Column and Row Affect Text Appearance

- A **Column** places Text elements vertically, so each text appears below the previous one.
- A **Row** places Text elements horizontally, so they appear side-by-side.
- The layout composable (Column/Row) controls structure and alignment, while Text controls content and styling.

3 Simple Wireframe Design

Below is a basic visual structure of the screen:

```
-----  
|      Title (Text)      |  
|    Subtitle (Text)    |  
|                        |  
| Paragraph text goes here. |  
| It explains something in |  
| detail.                |  
|                        |  
| [Button1] [Button2]    |  
-----
```

◆ Composable Mapping

- **Column**
 - Wraps entire screen content (vertical structure)
- **Text**
 - Title (large font size)
 - Subtitle (smaller font size)
 - Paragraph text
- **Row**
 - Wraps the two horizontally aligned text buttons

- **Text (or TextButton)**

- Button 1
- Button 2

Structure idea:

- Column
 - Text (Title)
 - Text (Subtitle)
 - Text (Paragraph)
 - Row
 - TextButton
 - TextButton

Reflection

Planning the text layout visually before coding helps in understanding the structure of the UI clearly. It prevents confusion while writing composables and reduces layout mistakes. Visual planning also makes spacing, alignment, and hierarchy easier to manage. Instead of randomly placing elements in code, we can build the UI logically and efficiently.